OpenADMET CYP Challenge — Inhibition Potency (Non-TDI) Track
Model Description Report

Team / leaderboard alias: Cheminfo
Submission: blind test set, 750 compounds
Predicted property: pIC50 of direct inhibition, for CYP1A2, CYP2C9, CYP2D6 and CYP3A4
Evaluation metric: ST-RAE (soft-threshold relative absolute error), macro-averaged over the four isoforms

1. The metric determines the model

ST-RAE is not an error metric in the usual sense, and every design decision below follows from its exact form:

```
soft_err_i     = max(0, y_i - conf_high_i) + max(0, conf_low_i - y_i)
baseline_err_i = max(0, mean(y) - conf_high_i) + max(0, conf_low_i - mean(y))
ST-RAE         = sum_i soft_err_i / sum_i baseline_err_i
```

A prediction falling inside the measured compound's credible interval `[conf_low, conf_high]` contributes **exactly zero**; outside it, only the distance to the nearer edge is charged. The denominator is the same quantity for the constant predictor `mean(y)`, so ST-RAE = 1.0 means "no better than predicting the mean". Two exact properties follow and were used throughout:

1. The denominator does not depend on our predictions, so a change in ST-RAE is **exactly linear** in the change in total soft error. Every post-processing step below could therefore be sized arithmetically before submission rather than tuned.
2. The derivative with respect to a uniform shift is a pure count, `#{ŷ > hi} − #{ŷ < lo}`.

The band widths are strongly potency-dependent — for CYP3A4 the median band is 2.493 log units in the weakest quintile against **0.135** in the most potent, and for CYP1A2 1.369 against 0.192. Tolerance is almost entirely at the weak end, so accuracy on potent compounds carries nearly all the cost.

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2. Training data

Only challenge-distributed data is used in the submitted columns. No external corpus contributes to any live column.

Dataset	Role	Size
`cyp-challenge-TRAIN_inhibition.csv`	Dose–response pIC50, primary labels	4,905 compounds
`cyp-challenge-single-concentration-TRAIN.csv`	Single-shot screen; auxiliary heads and pseudo-labelling	4,376 compounds × 4 isoforms
`cyp-challenge-TEST-BLINDED.csv`	Blind prediction set	750 compounds

Per-isoform dose–response label counts: CYP1A2 1,412 · CYP2C9 1,285 · CYP2D6 1,493 · CYP3A4 2,335.

> **External data was assembled and deliberately shelved.** A 30,980-row union corpus (6,141 challenge rows plus 24,839 external, carrying ChEMBL potency for 8,196–14,381 compounds per isoform

> and NCATS qHTS maximum-response for 15,602) was built and trained. It is **not** in the submitted file. The reason is an endpoint mismatch rather than a data-quality one: ChEMBL CYP potency carries a +0.45–0.65 log offset and a hard floor at 4.00 against this assay, so merging it into the scored targets is a truncation rather than a censoring, and it moved the CYP3A4 training mean by +0.96 while cutting the weak end from 51% to 8% of rows. Where external data was used in development it was carried on **separate heads**, never merged into a scored target.

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3. Structure preparation

Structures were prepared with **Schrödinger LigPrep** at **pH 7.4**, then audited for systematic protonation and bond-order artefacts and corrected by six RDKit scripts applied in a fixed order through a single driver, `clean_smiles_charges.py`. The driver executes only the function definitions from each script, carries molecules as RDKit objects with explicit hydrogens, and writes canonical SMILES to a new column. Structures LigPrep produced that could not be repaired locally (azolium, iminol, iminium and imine tautomers) fall back to the raw input SMILES.

#	Script	Correction
1	`neutralize_sulfonamide_N.py`	R-SO_2-[N^-]-R′ → R-SO_2-NH-R′
2	`neutralize_nplus.py`	Pyridinium-type [nH^+] → n; quaternary [n^+] left unchanged
3	`neutralize_morpholine_N.py`	Morpholine [NH^+] → N (pK_a ≈ 6.5, not dominant at pH 7.4)
4	`neutralize_iminium_N.py`	Amidine / guanidine / iminium C=[NH^+] → C=N, sp^2 nitrogen only
5	`fix_amide.py`	N=C([O^-]) → N-C(=O) with corrected bond orders
6	`fix_phenol.py`	c[O^-] → c-OH

The correction is material: 252 of 4,905 training rows and 52 of 750 test rows change, and the count of charged molecules falls from 941 to 718 in training and from 128 to 80 in the test set. Identical scripts are applied to both.

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4. Model architecture and training

4.1 Multitask ChemProp with single-shot auxiliary heads

A single ChemProp 2.2.3 D-MPNN trunk carries **eight regression heads**: the four scored `pIC50_direct_inhibition` endpoints at task weight 1.0, and four auxiliary single-shot `log2fc` heads at weight 0.2. RDKit 2-D descriptors are concatenated to the feed-forward network.

Setting	Value
Trunk	randomly initialised D-MPNN, hidden dimension 400, depth 4
FFN	3 layers; hidden dimensions cycled over (800, 600, 800, 400, 800) across members
Dropout / batch size	0.15 / 128
Max epochs / patience	400 / 50
Learning rate	1×10^{-4} → 1×10^{-3} → 1×10^{-4}, 5 warm-up epochs
Ensemble	**15 members** = 5 seeds (42, 123, 456, 789, 1337) × 3 split methods (random, stratified, scaffold), validation fraction 0.12

Censoring	dose–response readings at or below the top screening concentration (pIC50 4.305) are left-censored under `BoundedMSE`
Prediction clip	[1.5, 9.0]

The stratified split is stratified jointly on which isoforms are labelled and on activity quartile; the scaffold split is a Bemis–Murcko whole-scaffold hold-out.

> **A pretrained trunk was tested and rejected.** The CheMeleon foundation D-MPNN (hidden dimension 2048, depth 6, 8.7 M parameters) was trained as an alternative trunk. It wins on raw error but loses on ranking, and crossed with the band loss it read null-to-negative on all four isoforms. No submitted column uses it.

4.2 The band loss

Because ST-RAE charges nothing inside the credible interval, the training loss was changed to match. Per labelled cell, in scaled target space:

```
upper = y + hi
lower = y - lo
st    = relu(pred - upper) + relu(lower - pred)
L     = alpha * st + (1 - alpha) * (pred - y)**2
```

where `hi` and `lo` are that cell's own credible-interval half-widths, carried asymmetrically because 8.6 / 7.8 / 14.5 / 27.5% of CYP1A2 / 2C9 / 2D6 / 3A4 rows have one side more than 1.5× the other. At `alpha = 1` this term *is* the ST-RAE numerator exactly; the retained 30% of mean-squared error exists because a pure hinge has zero gradient inside the band and so cannot order compounds that are already in it.

α = 0.7 is used for every submitted band-loss column. The pure-hinge setting α = 1.0 was subsequently run in full and **refused on all eight acceptance gates** (four isoforms × two baselines), failing on the 90th-percentile error everywhere, so the parameter is settled at 0.7. Censoring composes with the band: the bounded-loss clamp is applied to the prediction first, then the band. ChemProp 2.2.3's loss signature has no slot for a per-cell band, so the bands are carried in the widened target matrix.

An alternative, inverse-band-width *sample weighting*, was measured and rejected: it is mathematically near-identical to 1/σ weighting (correlation 0.993–0.995 with band width).

4.3 The CYP3A4 column

CYP3A4 is served by an earlier, different architecture that no later run has beaten at the operating point. It pairs a primary 8-head model trained on real labels only with a **proxy model** trained only on compounds that were screened but never given a dose–response curve, using a fold-internal single-dose → pIC50 calibration. The final value is the primary model's, except where the proxy predicts weak activity, where the two are blended with a per-isoform weight chosen out of fold. Censoring at pIC50 4.305 and inverse credible-interval per-datapoint weighting are both applied. Twenty-two later local candidates were built to replace it and all twenty-two lost.

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5. Pseudo-labelling the single-shot screen

Only about one third of screened compounds received a dose–response curve, and for CYP2D6 the curves were run preferentially on compounds that already looked like hits. All 1,493 CYP2D6 pIC50 labels are therefore inhibitors, while 2,883 compounds that showed no activity have an empty cell — and the blind set is full of non-inhibitors. Training on the labelled subset alone places the whole CYP2D6 column far too potent.

The fix exploits a property of the curve-fitting software rather than of the enzyme. When a compound genuinely does nothing, the fitter returns a fallback value, and those values cluster tightly and agree across isoforms to within 0.04 log units (CYP1A2 2.168 · CYP2C9 2.333 · CYP2D6 2.168 · CYP3A4 2.206), which identifies them as a software floor. **Tier S** assigns each isoform's *own* floor value to every compound in the clearly-inactive zone of the single-shot screen — no fitted curve, no cross-isoform borrowing.

Tier	Rule	Typical error	Used?
S	Compound clearly did nothing → that isoform's floor value	0.15	**Yes**
A	Isoform has its own nearby data → calibrated curve	0.17–0.47	No
B	Borrowed from other isoforms	0.16–0.59	No
C	No data anywhere → left empty	—	No

Only tier S is used. Its contamination rate is 2.3% (one of 44 zone compounds that do have a real pIC50 sits above 3.0) with mean error 0.168, against 23% for the next tier down — ten times worse. In total **2,687 pseudo-label cells** are added, of which **2,424 are CYP2D6 non-inhibitors**, taking CYP2D6 from 1,493 to 3,917 labels.

> **The decisive detail: tier-S labels are point estimates, not censored upper bounds.** Real dose–response readings at or below pIC50 4.305 are left-censored, which is correct for a measurement that only establishes an upper bound. Applying the same treatment to 2,424 pseudo-labels all sitting at 2.168 would make "predict anything below 2.168" free, and with no gradient under the bound the cheapest joint solution is a constant — the CYP2D6 head collapses. Censoring is therefore enabled for real measurements and disabled for pseudo rows, and a hard abort guard fires if the CYP2D6 prediction standard deviation falls below 0.5. This is not a hypothetical failure mode: another entrant fed roughly 2,900 CYP2D6 non-inhibitors in as censored labels and watched σ collapse to 0.07.

Independent corroboration: moments recovered from the public leaderboard put the blind CYP2D6 mean at 3.11 with standard deviation 1.51; the pseudo-labels, built only from laboratory data, move the CYP2D6 training mean from 4.78 to 3.165 with standard deviation 1.391. Live, tier S took CYP2D6 ST-RAE from 0.7754 to 0.5806 and rank 45 to 19.

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6. Post-processing

6.1 The prediction floor

No compound on any isoform has a credible-interval upper bound below 3.10. A prediction below that is therefore **strictly dominated**: it can never earn band credit and can only lose it. Each column is floored at that isoform's minimum observed upper bound.

Isoform	Floor	Predictions raised	Mass moved (log units)
CYP1A2	3.1194	57	23.6
CYP2C9	3.2130	34	15.3
CYP2D6	3.1478	**440**	**293.1**
CYP3A4	3.1079	91	42.9

This is verified rather than argued: over the full cross product of every moved prediction against every labelled band, the maximum increase in soft error is +0.000000. Live, the CYP3A4 floor alone read −0.0176 against a pre-registered −0.010 to −0.018, and the CYP2D6 floor −0.0538 against a pre-registered −0.055. A standing audit now checks every column for sub-floor mass before submission, at a measured cost of 0.00041 ST-RAE per dominated log unit.

6.2 Affine placement and scaling

Predictions are placed on the blind-set moments recovered from the public leaderboard (CYP1A2 mean 4.32 / sd 1.39; CYP2C9 4.89 / 0.97; CYP3A4 4.74 / 1.15). The scale target is $\boldsymbol{\rho \cdot \sigma_y}$, where ρ is taken as √R² from the board — the attenuation-optimal scaling of a predictor of correlation ρ — *not* the recovered standard deviation itself, which would optimise calibration of spread rather than error. Three live probes establish the rule:

Probe	% of ρ·σ, before → after	Live Δ ST-RAE
CYP1A2 widen, step 1	81% → 100.7%	**−0.0080 won**
CYP1A2 widen, step 2 (a = 1.20)	100.7% → 120.9%	+0.0408 lost
CYP3A4 widen (a = 1.20)	109% → 131.2%	+0.0601 lost

Narrowing interacts with the floor and must be constructed around it. A pure affine narrowing of CYP3A4 to 100% of ρ·σ lifts the 91-compound floor pile from 3.1079 to 3.2524 — above the ≈3.15 band ceiling — converting previously free predictions into charged error at a modelled cost of +0.0054. The submitted construction instead **holds the pile at the floor** and rescales only the 659 compounds above it, re-solving the scale and offset from the full-column mean and standard deviation constraints. Because the pile is a tied block at the column minimum, holding it preserves ordering exactly (Spearman 1.000000), keeping the change a pure placement probe.

6.3 Order-only refinements

CYP2C9 additionally carries a QSAR rank perturbation: the live ChemProp ranking is mixed at weight 0.10 with the mean rank of a LightGBM model, a PaDEL/SVM model following Xu's protocol, and an MLP, then quantile-mapped back onto the live column's own empirical distribution, so only the ordering moves. CYP1A2 is a 23-member equal-weight consensus in which each member's blind column is z-scored within itself before averaging, members are deduplicated by ranking at ρ ≥ 0.999, and the result is affinely placed.

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7. Final per-isoform configuration

Isoform	Model source	Post-processing, in order
CYP1A2	23-member z-score consensus, blind-marginal arm	Placed on recovered moments (4.32 / 1.39) → floor 3.1194. No further affine.
CYP2C9	8-task band-loss ChemProp, tier S, α = 0.7	Floor 3.2130 → QSAR rank perturbation w = 0.10, quantile-mapped (order only)
CYP2D6	8-task band-loss ChemProp, tier S (2,424 pseudo cells), α = 0.7	Floor 3.1478 (440 rows). Held unchanged thereafter.
CYP3A4	Primary + proxy blend ensemble (Section 4.3)	Floor 3.1079 (91 rows, pile held) → floor-held attenuation-matched narrowing, scale 0.9145 on the 659 above the floor, mean 4.7967, sd 0.9612 = 103.7% of ρ·σ

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8. Submitted predictions and standing

The submitted file is `submission_3a4_narrow_floorheld_atten.csv`, 750 rows, validated with the challenge's own activity validator. Measured marginals: CYP1A2 4.4746 / 1.0779 · CYP2C9

4.8805 / 0.7977 · CYP2D6 3.6979 / 0.8386 · CYP3A4 4.7967 / 0.9612, with zero sub-floor predictions on any isoform.

Board	ST-RAE	Rank (field 273)
CYP1A2	0.5090	61
CYP2C9	0.4265	78
CYP2D6	0.5072	27
CYP3A4	0.5057	107
Overall (MA-ST-RAE)	**0.4871**	**63**

Read 2026-11-02. This is the best macro-average recorded on this track. The three unchanged isoform columns reproduced their previous scores to four decimal places, confirming that the CYP3A4 narrowing was the only variable in the entry.

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9. Software

ChemProp 2.2.3 (D-MPNN, `BoundedMSE`, custom band-loss metric); PyTorch Lightning; RDKit (descriptors, scaffold splitting, standardisation, the six charge-correction scripts); LightGBM, scikit-learn and PaDEL-Descriptor for the QSAR rank-perturbation arm; Schrödinger LigPrep for structure preparation. Ensemble training was run on an NVIDIA T4 GPU instance.